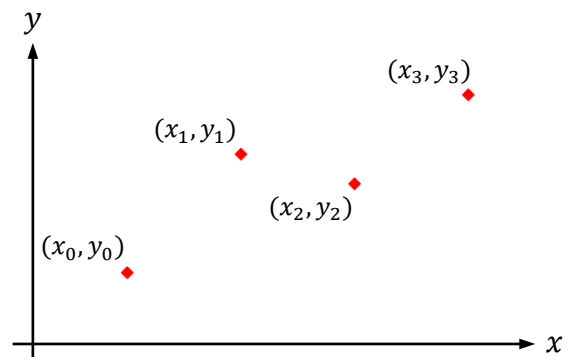
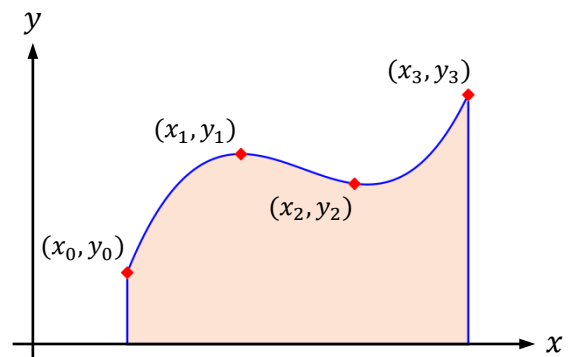


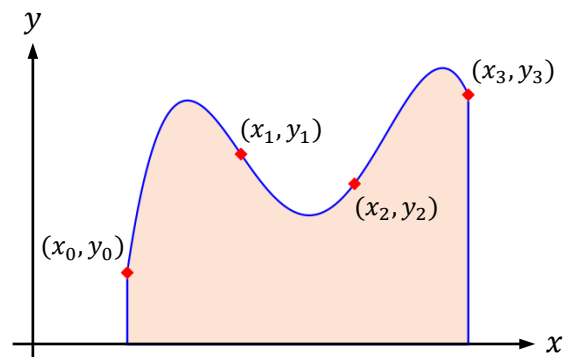
Concept & Outline



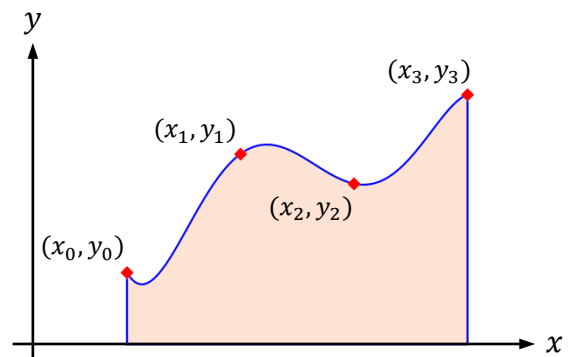
Concept & Outline



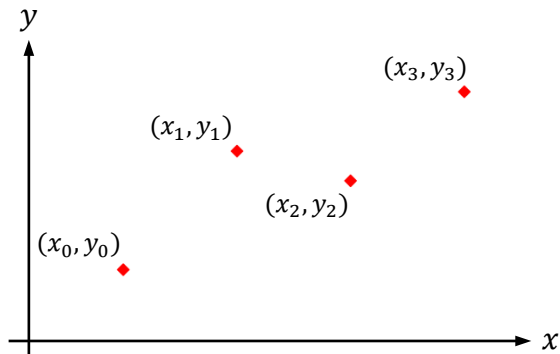
Concept & Outline



Concept & Outline



Concept & Outline



Integrand is known:

$$\int_{x_0}^{x_n} f(x) \cdot dx \cong \sum_{i=0}^k w_i \cdot f(\xi_i); \quad x_0 \leq \xi_i \leq x_n, n > 0, k > 0$$

A data set is known (integrand is not known):

$$\int_{x_0}^{x_n} f(x) \cdot dx \cong \sum_{i=0}^n w_i \cdot f(x_i); \quad n > 0$$

⑥ Numerical Integration

➤ Newton-Cotes Formulas of Closed Type

Interpolation based formulas.



❖ The Trapezoidal Rule

❖ Simpson's 1/3 Rule

❖ Simpson's 3/8 Rule

➤ Newton-Cotes Formulas of Open Type

Extrapolation based formulas.

❖ The Midpoint Rule

❖ The Trapezoidal Rule

❖ Milne's Rule

➤ Gauss-Quadrature Formulas

Relocation of Gauss points and their weighing factors for weighting functions.

❖ Domain Transformation

❖ Gauss-Legendre Quadrature

➤ Multiple Integrals

Applications of numerical integration formulas.

➤ Improper Integrals

Applications of Newton-Cotes formulas of open type.